

Viransh Shah

B.A.Sc. Computer Science — McMaster University 2026

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Highlights of Qualifications

- New graduate (June 2026) with 12 months of full-stack development co-op experience at a federal agency, seeking full-time software engineering or cybersecurity roles.
- Proficient in full-stack web and desktop development using Vue.js, React Native, TypeScript, Python Flask, and SQL with hands-on deployment experience on Azure and Linux.
- Strong analytical and problem-solving skills demonstrated through a 3.7 GPA (Dean's Honours List), competitive cybersecurity placements, and teaching assistant roles across three terms.

Education

McMaster University

Graduation June 2026

Bachelor of Applied Science in Computer Science, Minor in Statistics

Hamilton, Ontario

- Dean's Honours List: Maintained a GPA of 3.5 or higher (GPA: 3.7 / 4.0)
- **Databases (A+)**: ER Diagrams, SQL queries, Relational Data Models, Dependency Theory, and transaction Deadlocks.
- **Data Structures and Algorithms (A+)**: Stacks, Queues, Graphs, Trees, sorting algorithms, and hashing.
- **Computer Networks and Security (A+)**: TCP/IP, DNS, HTTP, routing, RTT analysis, and Wireshark-based packet analysis.
- **Information Security**: Applied cryptography including one-time pads, LFSRs, hash/collision analysis, and modular linear algebra.

Experience

Environment and Climate Change Canada & Canada Water Agency

Sept 2024 – Aug 2025

Full Stack Developer - IMWEBs Viewer & Nutri View

Burlington, Ontario & North York, Ontario

- Developed and extended the IMWEBs Viewer/Nutri View, a cross-platform desktop and web application for visualizing watershed and environmental data through interactive tables, graphs, and maps using Vue.js, Tauri, and Python Flask.
- Built data ingestion and conversion workflows for Excel-based BMPs, performance measures, and water quality data into SQLite, and integrated shapefiles and rasters into GeoPackages.
- Packaged the application for Windows, macOS, and Linux, deployed services on Windows and Azure VMs via IIS over HTTPS with a reverse proxy to Flask, and authored detailed technical documentation.
- GitHub: [ECCC-IMWEBs-Viewer](#) and [Nutri-View](#).

McMaster University

Jan–Apr 2024; Sep–Dec 2025; Jan–Apr 2026

Teaching Assistant - SFWRENG 2C03, SFWRENG 2XC3 & COMPSCI 1XD3

Hamilton, Ontario

- Led tutorials and 2-hour in-person labs on algorithms, data structures, and software development fundamentals across three terms.
- Supported students through lab exercises, assignments, and team projects with hands-on practice in Linux, Git, GitHub, and C programming.
- Guided students in COMPSCI 1XD3 to build and deploy websites using HTML, CSS, JavaScript, and PHP.
- Evaluated assignments, midterms, and final exams using grading rubrics and provided constructive feedback.

McMaster Cyber Society

Sept 2023 – June 2025 (MCP), July 2025 – Present (MCS)

Founding Executive Member

Hamilton, Ontario

- Competed in CTFs using Kali Linux, Nmap, Metasploit, and Wireshark for cybersecurity, cryptography, and reverse engineering challenges.
- Secured 2nd place as MacHackers Team in CyberSci 2025, representing McMaster ([photo](#)).
- Developed skills in incident response, log analysis, password cracking, binary exploitation, and attack chain analysis through TryHackMe, Advent of Cyber, and CyberStart Canada.

Projects

Slicc | *Bill Splitting and Restaurant Recommendation App* | [GitHub](#)

Sept 2025 – Apr 2026

- Cross-platform mobile app built with React Native, Expo, and TypeScript on Firebase (Firestore, Realtime Database, and Storage); currently usable on Android via APK.
- Implemented end-to-end encrypted group chat using AES-256-GCM with ECDH P-256 key exchange, and built the group scheduler and event creation features.
- Owned the restaurant scoring system that ranks venues by dietary, location, and price preferences to surface personalized matches as shareable location cards in group chats.
- Contributed to expense splitting workflows and collaborated in a 7-person Agile capstone team with iterative code reviews and continuous testing.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C, SQL, PHP, Bash, PowerShell, HTML/CSS, LaTeX

Frameworks & Libraries: Vue.js, React Native, Expo, Flask, Tailwind CSS, Electron, Tauri

Tools & Platforms: Git, GitHub, Docker, Azure, Google Cloud, SQLite, VirtualBox, GNU/Linux, IIS

Certificates: [Hackfinity Certificate of Completion](#), [Pre Security Certificate of Completion](#), [Cyber Security 101 Certificate of Completion](#)